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Early Triassic magnetic polarity time scale—integration of magnetostratigraphy, ammonite zonation and sequence stratigraphy from stratotype sections (Canadian Arctic Archipelago)

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ABSTRACT

Stratotypes defining the stages of the Early Triassic (Griesbachian, Dienerian, Smithian and Spathian) are located on Ellesmere and Axel Heiberg islands in the northern Canadian Arctic. Ammonite-rich horizons are within a clastic outer shelf-to-slope facies of thick progradational wedges of mudstones and siltstones. Three sections were sampled for magnetostratigraphy and interpreted for transgressive and regressive pulses of sedimentation. Using the ammonite zonation as a guide, the transgressive-regressive cycles and magnetostratigraphies have been correlated among the sections and to the published Triassic sequence stratigraphy time scale, thus enabling definition of the magnetic polarity pattern for the upper Griesbachian to Smithian stages in multiple sections. The magnetic polarity and associated sequence stratigraphy pattern for the lower Griesbachian and for the Spathian were derived from single sections. The Griesbachian and Dienerian stages each have two pairs of normal- and reversed-polarity chrons; the Smithian is predominantly of normal polarity, and the Spathian is predominantly of reversed polarity. This magnetic polarity time scale may help to resolve age correlations of North American redbed facies and to define the Permian–Triassic boundary.

After correction for variable structural orientations, the mean directions of magnetization from the three sites converge at 296° declination, 57° inclination ($k = 60$, $\alpha_{95} = 16.5°$; equivalent pole = 41° N, 161° E; paleolatitude = 38° N), which is consistent with the pole derived from nearby Early Permian volcanics and supports a postulated post-Early Triassic, pre-Tertiary counterclockwise rotation of this region with respect to cratonic North America.

1. Introduction

The magnetic polarity time scale for the late Mesozoic has been constructed through the correlation of magnetostratigraphies from fossiliferous sedimentary sections to the M-sequence pattern of marine magnetic anomalies (e.g., [1–8]). This polarity time scale for the Late Jurassic and Cretaceous has enabled high-resolution stratigraphic correlations and dating of sedimentary sequences within the Atlantic and Tethys basins, and recognition of geological stage boundaries (e.g., [4,6–11]).

The magnetic polarity time scale for the earlier portions of the Mesozoic has received much less attention, and for the Jurassic, magnetobiostratigraphic correlations have been verified for only a few stages (e.g., [12–16]). The Triassic presents a particular challenge to stratigraphers due to the subaerial exposure of the cratons; approximate ages of the continental sediment facies have been assigned from vertebrate and pollen zonations, but these biostratigraphic scales are of low resolution and lack detailed calibration to the standard ammonite and conodont zonations established in marine facies. Magnetostratigraphy for many of

these Triassic redbed formations is available (e.g., [17–27]); however, for most of the Triassic, a magnetic polarity time scale has not yet been developed for use in correlation and dating [28]. Lower Triassic shallow-marine sequences in China have yielded polarity patterns and associated tentative correlations to continental redbed facies [29–30]; however, faunal dating within these nearshore facies, the majority of which consist of dolomitic and oolitic carbonates and of marginal marine clastics, is not well defined [30]. Also, fossiliferous shallow-water carbonates of the Early and Middle Triassic in Nevada suffered pervasive remagnetization during the mid-Jurassic [31].

Early Triassic–Late Permian stratigraphy is additionally hindered by provincialism of marine fauna and by disagreements among paleontologists as to how to define the stage boundaries [32–38]. One major difficulty is the existence of a near-global hiatus between Permian and Triassic strata coinciding with the end-Paleozoic mass extinction—either the latest Permian, or the earliest Triassic, or both, may be absent. This problem is further exacerbated by the lack of agreement as to how to recognize or define the basal Triassic in different faunal provinces [38]. In order to standardize definitions of boundaries of geological time, thereby creating a consistent international nomenclature, the International Stratigraphic Commission has suggested that magnetostratigraphy should be a factor in selecting the stratotypes of such boundaries [39]. Towards this end, potential Permian–Triassic boundary stratotypes in China and Pakistan have successfully yielded polarity patterns [29,30], although age control in the uppermost Permian stage and above the P/T boundary are not well defined.

The Early Triassic is subdivided into four stages: Griesbachian, Dienerian, Smithian and Spathian. The "Early Triassic" of current usage is probably equivalent to the "Scythian" stage of Alpine stratigraphy and to the lowest of the three ("Trias") subdivisions of the classic Germanic Triassic. The four stages were first recognized and formally defined using a succession of ammonite horizons within a thick sequence of marine clastic sediments exposed in the Canadian Arctic Archipelago on Ellesmere Island and on Axel Heiberg Island [32,33]. We obtained magnetostratigraphies from three overlapping sections in this Arctic region. The sampling included the stratotype sections of the Griesbachian and Smithian stages and their component ammonite zones and also spanned strata and ammonite horizons equivalent to the nearby stratotype sections of the Dienerian and Spathian stages. The reproducible polarity patterns can also be correlated to upward shallowing sedimentary sequences [40], which may be a regional signature of Triassic global eustatic sea-level changes [41,42].

2. Stratigraphy and depositional sequences

The Sverdrup Basin of the northern Canadian Arctic was a depositional basin through most of the Paleozoic and Triassic. Mapping of the thick Triassic successions of clastic shelf to basinal sediments [43,44] led to the discovery of ammonite-rich horizons of Early Triassic age within the more basinward facies. Creeks on northern Ellesmere and Axel Heiberg islands which contain the best exposures were named after famous paleontologists of the Triassic; in turn, the names of the four stages proposed for the Early Triassic refer to these "stratotype" exposures [32,33]. These four stages have become accepted internationally, although the ammonite zone subdivisions and the representative nature of the original Arctic stratotypes are still debated. Problems with these Arctic localities include provincialism of some of the index ammonites used to identify zones, apparent lack of adequate index fossils needed for calibration to standard conodont biozones (W. Sweet, written commun., 1987), and the sporadic nature of the occurrence of ammonite-rich horizons within the sections. Even so, these Canadian Arctic sections are one of the few places in the world containing a continuous and fossiliferous marine record of the Early Triassic.

In the ammonite zonation of the Arctic, the base of the Triassic is defined as the base of the Griesbachian stage which, in turn, is defined by the first appearance of *Otoceras* ammonoids [33,38]. However, some paleontologists, especially those specializing in conodonts, would prefer to define the base of the Triassic at a major evolutionary event corresponding to the middle of the Arctic Griesbachian stage [35,37]. This issue of an international boundary standard, which is compounded by the extreme rarity of "Lower Gries-

bachian" sediments [38], remains a matter for active debate by the paleontologists and the Triassic Subcommission. The terminology used in this paper will conform to the definitions originally proposed in these Arctic sections for the base of the Triassic and for the boundaries of the stages [33].

The oldest *Otoceras* assemblage (*Otoceras concavum* zone), hence the oldest documented portion of the Griesbachian stage in the Arctic, occurs at Griesbach Creek on Axel Heiberg Island (section GRC) [32,33]. This basal Triassic interval is represented by a minor upward coarsening succession. This ammonite zone and associated depositional sequence was not identified within the stratigraphic sections on Ellesmere Island.

The stratigraphy of the Lower Triassic of the Sverdrup Basin has been described and formalized by Tozer [43,45] and Embry [46]. In the region of the "type sections", the basal Triassic is disconformable upon Permian black radiolarian cherts. The Lower Triassic is represented by mudstones and siltstones of the Blind Fiord Formation, which has been interpreted as an outer shelf to slope-fan depositional environment [40,46]. At the margins of this basin, the sandstone-dominated Bjorne Formation was deposited. The maximum total thickness of the Lower Triassic Blind Fiord Formation is approximately 1000 m (section MB). This formation is subdivided into three members: Confederation Point Member, Smith Creek Member and Svartfjeld Member. Our sampling program included the stratotype sections (our sections MB and TZ) for each of these three lithologic members (Figure 1).

The threefold subdivision of the Blind Fiord Formation reflects the occurrence of three major transgressive-regressive cycles during the Early Triassic [40,46]. The lower two members, Confederation Point and Smith Creek, consist of medium-to-dark gray silty shale with thin siltstone interbeds. In the upper portion of these members, siltstone beds become thicker and more abundant, with some siltstone-rich intervals displaying synsedimentary slumping. The tops of the Confederation Point and Smith Creek members are defined by the tops of prominent siltstone-rich intervals underlying a sharp transition to shale-dominated facies. Towards the center of the depositional basin on Axel Heiberg Island, the thickness of each

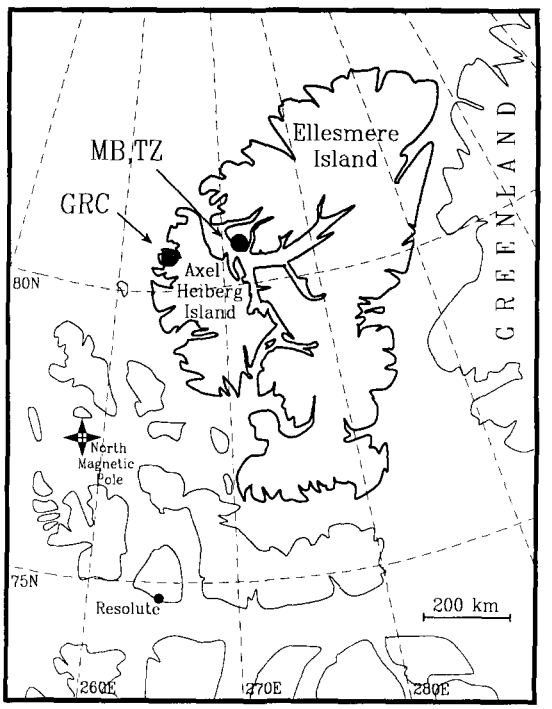


Fig. 1. Location map of magnetostratigraphic sections on the Canadian Arctic Archipelago. Sections MB and TZ are on Ellesmere Island, northwest of the junction between Nansen Sound and Hare Fjord. Section GRC is on Axel Heiberg Island, south of Bunde Fjord.

member decreases (e.g., the Confederation Point Member in our section GRC is only one-third of its thickness in section MB). The siltstone-rich upper portions of these two members correlate with prograding sandstone wedges on the basin margins [46]. Within each member is also a less-pronounced upward coarsening succession, terminating at an ammonite-rich condensation horizon.

The highest member, the Svartfjeld, consists of black silty shale with orange-weathering calcareous siltstone beds spaced at intervals of about 10 m, with an increasing abundance of these beds in the upper portion of the member. The Svartfjeld Member is abruptly overlain by Middle Triassic (Anisian) black, phosphatic, calcareous shale.

Section MB, along an unnamed creek 2 km north of Smith Creek on the Svartfjeld Peninsula of northwestern Ellesmere Island (80.6° N, 272.4° E) (Fig. 1), contains the stratotype sections of the Confederation Point Member and the Smith Creek Member [46]. We shall refer to the location of section MB as the "Creek of Embry" in the

stratigraphic figures. Nearly the complete Svartfjeld Member is exposed and was sampled along another creek about 0.5 km further north. Strata in this 1062 m composite section have an average dip of 45° towards 325°, with minor small-scale irregularities. Each ammonite zone of the Smithian stage can be recognized in beds which are lithologically correlated to similar fossiliferous horizons of the nearby "stratotype" of the Smithian stage (section TZ) studied by Tozer [33]. Age control for the remainder of the section is based upon the lithostratigraphic framework of Embry [40,46].

Section TZ is along Smith Creek, 2 km south of the MB section. It contains the stratotype of the Smithian stage [33], and extends upward into the type section of the Svartfjeld Member and downward into the Confederation Point Member. Structural orientation is similar to MB. Four major siltstone pulses or ammonite-bearing horizons within the 220 m section TZ can be lithologically correlated with similar unique beds within nearby section MB. From these correlations, it is apparent that most of section TZ has an average sedimentation rate of 65% of that of section MB.

Section GRC was sampled along Griesbach Creek on northwestern Axel Heiberg Island (80.5° N, 265.5° E), and contains the stratotype of the Griesbachian stage [33]. The 200 m section extends through the entire Confederation Point Member into the lower portion of the Smith Creek Member. Sampling was discontinued about 50 m below an igneous sill within the Smith Creek Member. The lithostratigraphic members here are approximately one-third of their thickness in section MB; these slower sedimentation rates are reflected in the oxidation of several silt-rich beds, in concentrations of ammonites within some shale intervals, in the increased concentrations of magnetic material, and possibly by the greater variety of lithologies. This relative condensation is consistent with the more basinal setting of section GRC. Structural attitudes vary from vertical towards N45° E in the lower Griesbachian to 70° towards N45° E in the Smith Creek Member. Ammonites are particularly abundant within all four zones of the Griesbachian, and have been reported from the overlying Dienerian [33].

The ammonite biozonation and associated definitions of Early Triassic stages in these sections is formalized in the North American standard zonation developed by Tozer and Silberling [33,47] and correlated to the standard Tethyan ammonite zonation [34,48]. Conodonts obtained by us from these Arctic sections were not diagnostic [W. Sweet, written commun., 1987]. Unfortunately, only about two or three ammonite-rich horizons occur within each lithostratigraphic member. Therefore, the "boundaries" between ammonite zones, hence between stages of the Lower Triassic, have an uncertainty corresponding to the spacing between fossiliferous horizons, which in the worst case is 200 m.

Identification of depositional sequences in seismic reflection transects and in facies transitions in outcrops of marine strata provides a tool for stratigraphic correlation [41,42,49]. A depositional sequence is a cycle of sea-level change beginning with a relative sea-level fall, expressed as a basinal shift of the coastal onlap pattern. In outcrop exposures of a clastic margin facies, this sequence boundary changes character from landward sections, where it is commonly an exposure surface, to basinward profiles, where it can occur as a conformable level within an overall shoaling upward succession with a change from late highstand facies of interbedded progradational deposits to lowstand facies of more massive aggradational deposits [42,50,51]. During sealevel lowstands, a coarsening upward wedge or fan will develop in basinal settings. Lowstand slope fans consist of stacked leveed-channel complexes and slump or debris-flow deposits [52]. The most evident sea-level change in exposures of clastic shelf facies is the condensation interval and succeeding facies change caused by transgression on the shelf and resulting starvation of clastic delivery to the outer shelf and basin [42]. Between this surface and the following attainment of maximum flooding on the shelf, the clastic input starvation produces a condensed section, commonly with concentration of pelagic fossils and biogenic carbonate, glauconite layers or a hardground caused by lithification [53]. The relative duration of sediment starvation within a condensed section increases basinward [42,50,51]. Basinal and outer shelf sedimentation during prolonged sealevel highstands consists of a coarsening upward succession, possibly with an organic-rich clayey interval at the base.

The northern Ellesmere to Axel Heiberg Island

transect of our paleomagnetic sampling sites represents an outer shelf to basinal portion of a clastic-dominated margin [40], and the facies successions were interpreted according to the models of depositional sequences for these settings. The boundaries between lithologic members in these sections are defined by a sharp upward transition from silt- and sand-dominated beds to clay-dominated strata, and therefore correspond to the beginnings of transgressive and highstand episodes [40]. Fossiliferous hardgrounds within the Confederation Point and Smith Creek members, associated with a change from underlying coarser grained to overlying clayey deposits, also indicate transgressive events. Transgressive and highstand deposits are represented by the clay-rich intervals; regressive and lowstand deposits are represented by an increasing abundance of siltstone beds, commonly with slumps and olistostromes in the upper portions. The placement of sequence boundaries in these settings is slightly subjective, in that there is a transition zone between clayey highstand sediments and silt-rich lowstand sediments. The response of a clastic shelf-basinal environment to fluctuating relative sea level results in highly variable rate of sedimentation, which, in turn, causes a distortion of the thickness of magnetostratigraphic or biostratigraphic zones relative to linear time. In general, the lowstand deposits dominate the sedimentation in the more basinal settings.

The ammonite-biostratigraphic ages of the three main progradational clastic wedges (the three lithostratigraphic members) of these Arctic sections correspond closely to the cited ages of the three Early Triassic episodes of eustatic sea-level change and coastal onlap of the sequence chronostratigraphy developed by Haq et al. [41]. Their sequence stratigraphy is derived largely from the Salt Range of Pakistan and from Spitsbergen and Bjornoya (Bear Island) in the European Arctic [42]. In the Haq et al. nomenclature for the Triassic sequence chronostratigraphy, these episodes are the 3rd-order cycles "1.2", "1.3" and "1.4" of the "UAA-1" 2nd-order supercycle within the "Upper Absaroka" 1st-order megacycle of the Phanerozoic. The earliest Triassic sequence "UAA-1.2" is further subdivided into 4th-order cycles of "1.2.1" and "1.2.2" [42]. However, it cannot be known whether these depositional sequences are exactly correlative until detailed magnetostratigraphy is available from these Salt Range and European Arctic sections.

We have extended the 4th-order cycle nomenclature downward through the addition of a basal subsequence "UAA-1.2.0" of earliest Griesbachian age, as identified in the Griesbachian stratotype. Strata of earliest Griesbachian (*Otoceras concavum* zone) are absent in the sections incorporated in the Haq et al. scale [40,42]. This earliest subcycle of the Triassic probably represents a fluctuation of sea level about a general extreme lowstand average, explaining the continued absence of sediment accumulation in the slope to outer shelf settings of the Ellesmere Island sections and of the sections incorporated in the Haq et al. sequence chronostratigraphy.

Details of lithostratigraphy, ammonite zonation and interpreted depositional sequence stratigraphy for each section are shown on the magnetostratigraphic charts (Fig. 5).

During the Cretaceous, especially from Hauterivian to early Cenomanian, the Sverdrup Basin was affected by episodes of basaltic volcanism and intrusions, which may coincide with the main rifting phase of the adjacent oceanic Canadian Basin [54–57]. The magnetostratigraphic sections were selected to avoid sills or dike swarms associated with this regional igneous activity. During the early Tertiary, the Sverdrup Basin experienced the "Eurekan orogen" pulse of tectonic disruption, which may have been associated with the openings of Baffin Bay and of the Arctic Ocean [54,58,59]. Uplift, block tilting, thrust faulting, halotectonic structures and horizontal displacements are common, but were not accompanied by metamorphism.

3. Analytical procedures and magnetic properties

A total of 342 samples of 2.5 cm diameter were cored from the three stratigraphic sections (195 at MB, 64 at TZ, and 83 at GRC). The density of sampling depended upon the sedimentation rates, but averaged twenty samples spaced stratigraphically within each ammonite zone of each section.

Even though the sites are within 10° of the present Magnetic North Pole (Fig. 1), if one removed all metal objects from the area, it was possible to accurately measure the azimuthal orientation of the cores and the strike of bedding

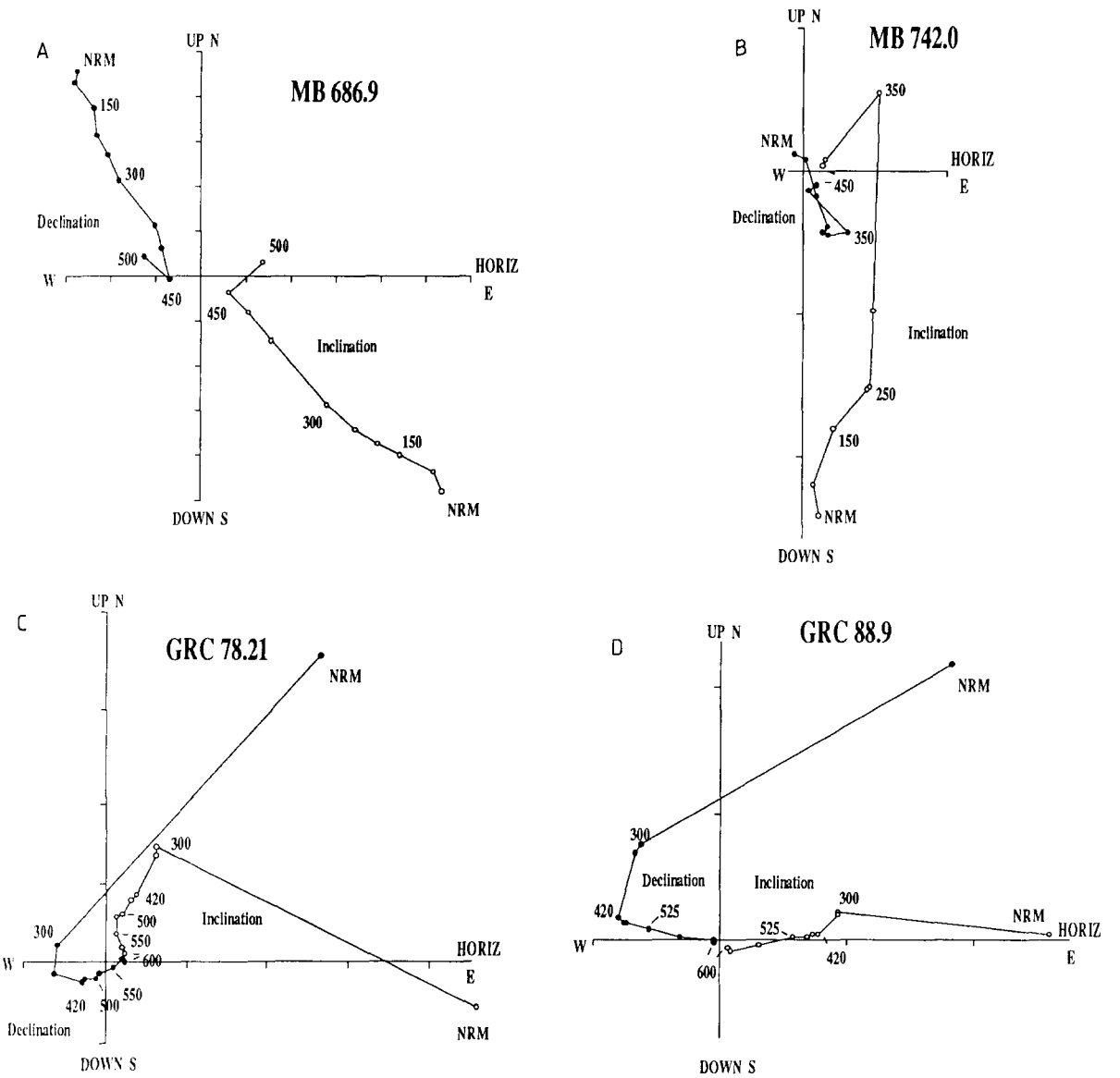


Fig. 2. Vector plots of thermal demagnetization behavior of typical samples. Declination and inclination (dots and circles, respectively) vectors correspond to the horizontal component and to the actual dip of the magnetization vector; vectors have been corrected for tilt of strata. (A) Sample MB 686.9 (Creek of Embry, late Smithian age, clayey siltstone); characteristic direction (300°–450°C; linear, but not to origin) = 333.7° decl., 46.8° incl. (normal polarity *N*) has a mean intensity of 1.0×10^{-4} A/m. Viscous behavior displayed at 500°C and above (not shown). One scale division = 5×10^{-5} A/m. (B) Sample MB 742.0 (Creek of Embry, Spathian age, clayey siltstone); characteristic direction (350°–450°C; quasi-linear, towards origin) = 142.2° decl., −50.4° incl. (reversed polarity *R*) has a mean intensity of 1.8×10^{-4} A/m. Viscous behavior displayed at 500°C and above (not shown). One scale division = 5×10^{-5} A/m. (C) Sample GRC 78.21 (Griesbach Creek, late Griesbachian age, greenish siltstone); characteristic direction (average of 570° and 600°C) = 87.1° decl., −30.4° incl. (reversed polarity *RP*) has a mean intensity of 1.4×10^{-3} A/m. One scale division = 5×10^{-3} A/m. (D) Sample GRC 88.9 (Griesbach Creek, Late Griesbachian age, greenish-red siltstone); characteristic direction (average of 570° and 600°C) = 272.4° decl., 42.4° incl. (normal polarity *NP*) has a mean intensity of 5.3×10^{-3} A/m. One scale division = 5×10^{-2} A/m.

by using an International Brunton Compass. Declination corrections, which averaged about 105° W and had insignificant daily variation, were determined twice-daily by multiple sightings on distant prominent landmarks recorded on oriented aerial photographs and topographic maps.

Paleomagnetic measurement and demagnetization were performed in a steel-shielded room at the University of Wyoming using a computer-linked two-axis ScT cryogenic magnetometer. In order to identify appropriate demagnetization procedures for each lithology, pilot suites from each section were treated by progressive thermal demagnetization in 50°C steps up to 630°C or by alternating field (AF) demagnetization up to 80 mT. Progressive thermal demagnetization was generally more effective for distinguishing the components of magnetization; therefore, the samples were demagnetized at 30°–40°C increments for five to nine steps through the temperature range of the characteristic magnetization.

Dark gray siltstones or silty claystones comprising the MB and TZ sections displayed steep-downward directions of initial natural remanent magnetizations (NRM) prior to bedding correction. NRM intensities averaged 2×10^{-4} A/m. This steep-downward component of magnetization was removed upon thermal demagnetization through 200° or 250°C, and this component is interpreted as an overprint of the present geomagnetic field or of the Holocene dipole field. Normal- or reversed-polarity directions were generally apparent upon heating to 300°C, with progressive removal of apparent characteristic directions continuing through 400° or 450°C (Fig. 2A and 2B). In contrast, progressive AF demagnetization up to 80 mT failed to yield antipodal directions which would indicate polarity. The intensities of these characteristic magnetizations averaged 3×10^{-3} A/m. At 500°C and above, samples generally displayed erratic directions, presumably the result of highly susceptible magnetic minerals having been formed at high temperatures. Similar magnetic behavior was displayed by some of the GRC sample set. Based upon the behavior during demagnetization, magnetite may be the main carrier of characteristic magnetization in the clayey siltstones of sections MB and TZ. Removal of a present-day overprint upon thermal demagnetization to 200° C, in contrast to failure of AF demagnetization at high levels to remove overprints, suggests that a high coercitivity mineral, such as goethite, is an important carrier of secondary magnetization. Alteration of this mineral at higher temperatures may explain the viscous or spurious directions displayed upon heating at 500° C and higher.

Within section GRC, there was considerable variation in demagnetization behavior among the various lithologies, but the most common features can be summarized as follows: Most samples displayed relatively strong magnetizations, which were more resistant to thermal demagnetization than the MB and TZ sets. The demagnetization procedure required closely spaced heating steps at higher temperatures in order to separate the characteristic directions (Fig. 2C and 2D). NRM intensities averaged about 1×10^{-2} A/m, but intensity variations ranged over two orders of magnitude among the different lithologies. A steep-downward component (prior to bedding correction), most likely an overprint of the present or Holocene field, was removed upon heating to 300°C. At the 340°C step, the majority of samples exhibited an unusual direction of magnetization directed upwards towards approximately 270° declination, −55° inclination after bedding corrections (or about 310°, −45° before bedding corrections). The origin of this component of magnetization is unexplained. This component was generally removed upon thermal demagnetization to 450° or 500°C to reveal clusters of antipodal directions stable through 525°C or higher. The average intensity of these apparent characteristic directions was 5×10^{-4} A/m. In general, those samples having a greater amount of clay displayed a greater tendency for development of erratic directions at higher temperatures. Samples which display directions stable to 600° C are generally reddish colored, suggesting that hematite, perhaps formed during early near-surface oxidation, may also be present and a carrier of characteristic magnetization.

Most samples of these three sections either attained a direction which remained stable through multiple heating steps as the intensity of magnetization decreased (i.e., univectoral decay to the origin on the vector diagrams), or displayed development of a semi-stable direction and intensity prior to onset of erratic magnetization at higher

temperatures. Characteristic magnetization directions and associated variances were computed for each sample by applying a least-squares three-dimensional line-fit (principal component analysis) to those sets of vectors in the plots that displayed removal of a single component, hence a linear path to the demagnetization path [60]. The intensity of characteristic magnetization was computed as the mean of the intensities of those vectors used in the least-squares fit.

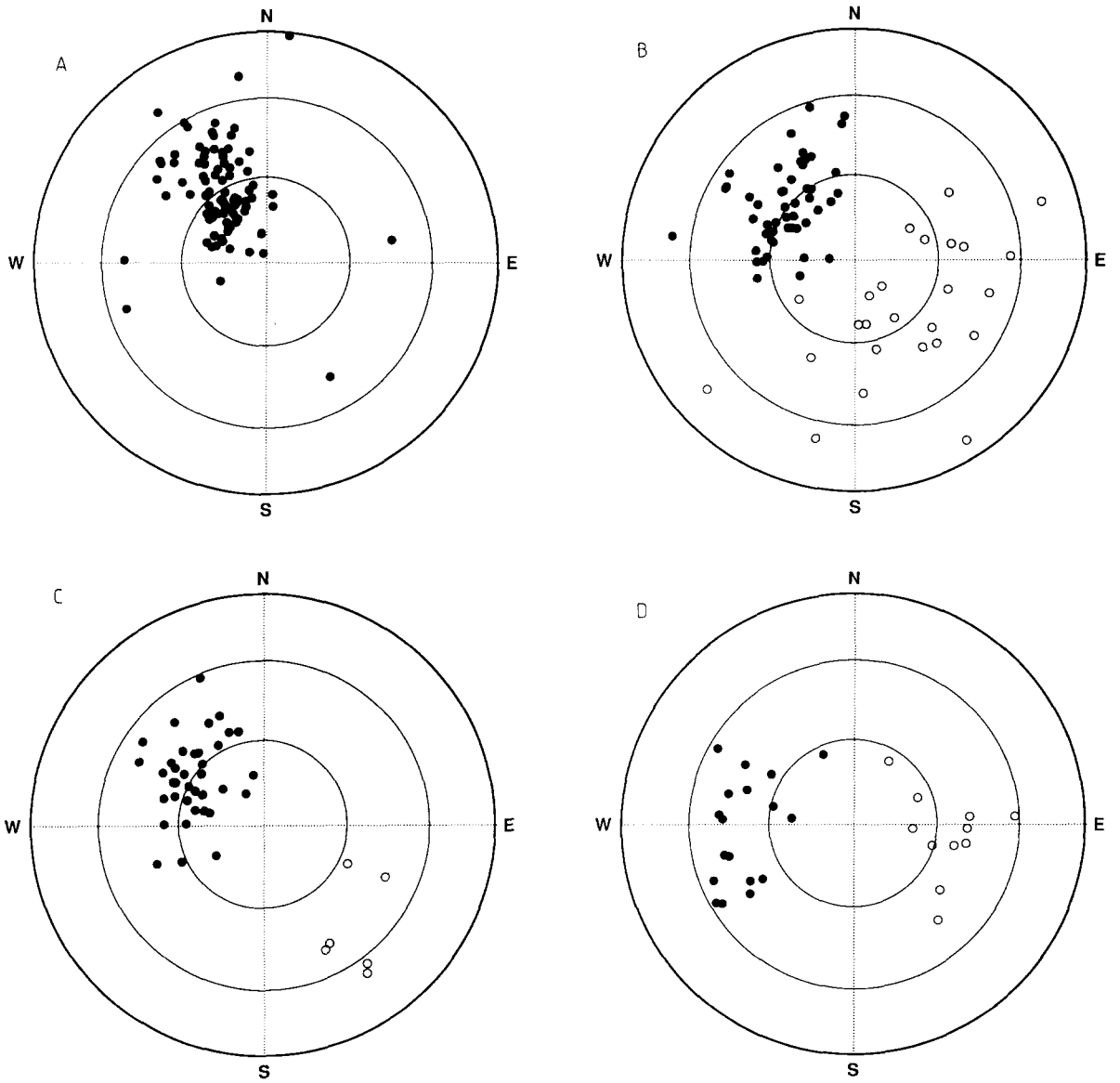


Fig. 3. Equal-area plots of directions of magnetization; dots are positive inclinations, circles are negative inclinations; all directions have been corrected for bedding tilt. In the plots of characteristic directions, samples with uncertain or questionable polarity (*INT, N?, R?*) or with poor directional reliability (*NPP, RPP*) have been omitted. (A) Directions of natural remanent magnetization (NRM) from the lower portion of section MB prior to demagnetization treatments. (B) Characteristic directions of magnetization from section MB. (C) Characteristic directions of magnetization from section TZ. (D) Characteristic directions of magnetization from section GRC. (Directions in (B)–(D) are used in VGP.)

Each characteristic direction was assigned a rating indicating its degree of precision (number of vectors incorporated in the least-squares fit, their colinearity, and presence of univectorial decay) and the reliability of the assignment of polarity:

(1) The quality of the characteristic direction was considered to be excellent and denoted by *N* or *R* in the stratigraphic plots only if the region of linear decay was represented by at least three demagnetization steps having a line-fit variance of less than 10° and preferably with a stable direction exhibited by decay toward the origin on vector diagrams. Only about 10% of the samples met these criteria.

(2) A characteristic direction was considered to adequately reflect the Triassic magnetic field direction and was denoted by *NP* or *RP* in those cases where either the polarity of the sample was easily interpreted, but where the standard deviation of the removed vector was greater than 10°, or where less than three demagnetization steps were used in the least-squares computation. Approximately one-third of the MB and TZ samples were of this quality. Such directions were assigned only half-weight in the computation of site mean directions.

(3) A characteristic direction was considered to be "poor" and denoted by *NPP* or *RPP* in cases where the polarity was clearly evident, but where the direction prior to onset of erratic behavior was not duplicated in lower demagnetization step. A "best step" was selected for these samples. The majority of samples from GRC were given this poor rating. Results from samples having "poor"-rated characteristic directions were omitted from computation of mean directions of the sites.

(4) Samples having indeterminate, intermediate or questionable polarity interpretations, denoted by *N?*, *R?* or *INT* in the stratigraphic plots, were omitted from the computation of mean directions for each site.

4. Mean directions

Directions of characteristic magnetization from each site are displayed in Fig. 3 for all samples rated *N*, *NP*, *R* and *RP*. Mean directions of these characteristic magnetizations and corresponding apparent poles of each site were calculated separately for normal-polarity and for reversed-polarity samples using a method modified from Fisher [61]. Vectors rated *NP* or *RP* were assigned half-weight. After the first iteration, sample directions falling outside two standard deviations from the initial mean were omitted. The intensity of this site mean direction was computed as the logarithmic average of the incorporated sample intensities, therefore allowing determination of a mean directional *vector* for each polarity. The vector sum of these mean normal-polarity and reversed-polarity vectors yields a more accurate representation of the mean characteristic direction, because any secondary magnetic vector which affects samples of each polarity equally will be removed by such a vector addition. However, for comparison, a separate mean characteristic direction was also computed by combining the normal-polarity and inverted reversed-polarity directions.

Composite mean characteristic directions for the Griesbachian–Dienerian, for the Smithian–Spathian, and for the combined Early Triassic stages were computed from the mean vector sums of the sites which contained strata of these ages. For TZ, where the vector sum is distorted due to the small number of reversed-polarity samples, the combined sample direction was incorporated in all-site computations. The mean apparent poles for the different age intervals are displayed in Fig. 4, and compared to other Mesozoic poles reported from Ellesmere and Axel Heiberg islands and to coeval Mesozoic poles from cratonic North America.

Several observations regarding the mean directions can be made:

(1) For all outcrops and all time intervals, normal-polarity and reversed-polarity samples display mean directions that are antipodal to within a few degrees. Even the worst case of section TZ, in which the number of reversed-polarity samples was inadequate for such a test, directions antipodal to within 20° are displayed. The antipodal nature of the magnetization directions supports the interpretation that the magnetic polarity zones represent Triassic magnetization set near the time of deposition. In addition, the mean intensities of normal-polarity and of reversed-polarity samples at each MB and GRC are nearly identical; hence, the mean direction *vectors* of both polarities are each *antipodal and equal*, implying that these mean

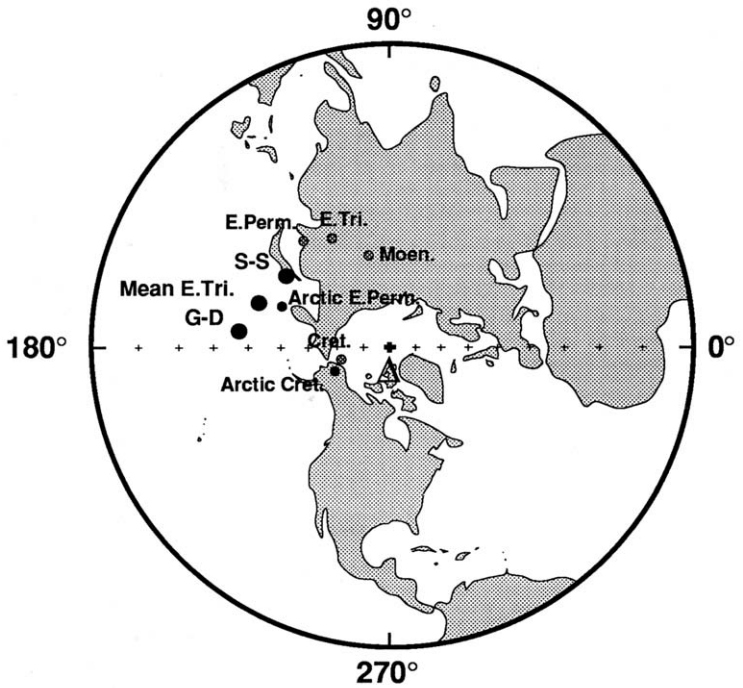


Fig. 4. Mean poles (VGP's and α_{95}'s) from the combined Canadian Arctic sites (Table 1; $G-D$ = Griesbachian–Dienerian age set, $\alpha_{95} = 17°$; $S-S$ = Smithian–Spathian age set, $\alpha_{95} = 25°$; *Mean E.Tri.* = entire suite combined, $\alpha_{95} = 19°$). Other Mesozoic poles from volcanics in this region are displayed as heavy dots: Early Permian (159.2° E, 48.1° N, $\alpha_{95} = 6.1°$) [67] and Cretaceous (203° E, 67° N, $\alpha_{95} = 6.6°$) [55] age. Coeval poles from North America are displayed as gray dots: *E.Tri.* = mean cratonic Early Triassic (118.3° E, 46.0° N, α_{95} not computed) [63], *Moen.* = Lower Triassic (mainly Smithian–Spathian stages) Moenkopi Formation (103° E, 55° N, $dp/dm = 1.6°/3.2°$) [19] from the Colorado Plateau which has rotated clockwise with respect to cratonic North America [63–65]. Early Permian (129° E, 41° N, $\alpha_{95} = 8°$) [67] and Cretaceous (196° E, 71° N, $\alpha_{95} = 4.9°$) [68] are also shown as gray dots. Ellesmere and Axel Heiberg islands (triangle) have experienced significant counterclockwise rotation relative to cratonic North America.

directions do not appear to be biased by secondary overprints.

(2) At individual sites, the fairly uniform bedding prevents a statistically significant fold test. Section GRC, which displays intrasite variation in bedding vectors of about 40°, has a dispersion parameter k for the uncorrected direction of 9.7 vs. a value of 15.9 after bedding correction, indicating an improvement in the clustering of characteristic directions upon correction for folding. An intersite fold test is available in the comparison of directions from MS and GRC, which have nearly a 90° difference in their average beddings. Their mean directions of characteristic magnetization after bedding correction converges to within about 10°, in contrast to the approximately 80° of separation before structural correction.

(3) Between the early Early Triassic (Griesbachian–Dienerian stages) and the late Early Triassic (Smithian–Spathian) there appears to have

Notes to Table 1: N = effective number of samples (or sites) used in calculation (characteristic directions of poor stability given half-weight). N_s = number of selected samples (questionable and indeterminant polarity samples omitted). ($N = Ns$ for "no weighting" calculation). α_{95} = radius of circle of 95% confidence about mean direction. K = dispersion parameter. Mean intensity = logarithmic mean intensity of selected sample set. VGP = virtual geomagnetic pole (dp/dm = oval of 95% confidence). (All-site mean VGP's are computed independently using site VGP set). N and R = normal- or reversed-polarity sample set, respectively. *Comb.* = Fisher statistics on all samples combined. $N + R$ = vector sum of mean normal- and reversed-polarity direction-intensity vectors. ** = site mean direction selected for all-site mean.

TABLE 1

Mean directions of characteristic magnetization (after structural correction)

	N/N_s	Mean direction				Mean intensity (10^{-5} A/m)	VGP			Lat. (°N)
		Decl. (°)	Incl. (°)	α_{95} (°)	K		Lat. (°N)	Long. (°E)	(dp/dm) (°)	
Unnamed creek (MB)										
Lower half = Griesbachian – Dienerian [total samples = 97]										
N	11 /17	294.8	52.0	10.0	22	3.6	36.1	163.8	(9.3/13.6)	32.6
R	7.5/14	284.3	60.3	23.8	7	3.7	42.9	176.8	(27.7/36.1)	41.2
Comb.	18.5/31	291.3	55.3	10.3	12	3.6	38.7	168.2	(10.4/14.6)	35.8
N + R **	18.5/31	290.1	56.3			3.6	39.6	169.7		36.9
Upper half = Smithian – Spathian [total samples = 97]										
N	20.5/32	313.9	60.4	6.2	28	4.8	47.5	145.8	(7.1/9.3)	41.3
R	6 /10	312.6	53.7	24.5	8	2.5	42.5	123.7	(23.8/34.1)	34.2
Comb.	26.5/42	318.5	59.2	6.7	18	4.1	46.8	140.4	(7.5/10.0)	40.0
N + R **	26.5/42	321.1	58.4			3.6	46.2	137.3		39.1
Entire Early Triassic — weighted [total samples = 194]										
N	30.5/48	306.0	58.4	5.1	27	4.4	44.2	153.7	(5.6/7.6)	39.1
R	13.5/23	308.2	59.6	16.9	7	3.1	45.9	151.7	(19.1/25.4)	40.4
Comb.	44 /71	306.6	58.8	5.8	15	3.9	44.7	153.2	(6.5/8.7)	39.5
N + R **	44 /71	306.9	58.9			3.8	44.9	152.9		39.7
Entire Early Triassic — no weighting										
N	48	309.8	57.9	4.5	22	4.1	44.2	149.6	(4.9/6.7)	38.5
R	23	313.2	61.4	11.9	8	3.1	48.6	146.9	(14.1/18.3)	42.5
Comb.	71	311.0	59.0	4.7	14	3.7	45.5	148.8	(5.3/7.1)	39.7
N + R	71	311.2	59.4			3.6	46.1	148.5		40.2
Smith Creek (TZ)										
Smithian – weighted [total samples = 64]										
N	25.5/35	303.3	58.7	5.2	31	6.8	44.2	156.7	(5.8/7.8)	39.5
R	4 /6	322.4	39.9	19.9	22	2.6	30.1	133.1	(14.3/23.8)	22.6
Comb. **	29.5/41	306.8	56.4	5.5	24	6.0	42.3	152.3	(5.7/7.9)	37.0
N + R	29.5/41	310.1	54.0			4.6	40.3	148.3		34.5
Smithian — no weighting										
N	36	302.5	59.1	4.8	26	6.3	44.5.	157.7	(5.4/7.2)	40.0
R	6	319.0	39.6	15.1	21	2.9	29.4	136.8	(10.8/18.1)	22.4
Comb.	42	305.8	56.5	5.0	21	5.6	42.2	153.5	(5.2/7.2)	37.1
N + R	42	309.2	53.3	4.5			39.5	149.1		33.8
Griesbach Creek (GRC)										
Griesbachian – Dienerian — weighted [total samples = 83]										
N	10.5/18	268.5	50.2	12.3	15	41	30.2	182.6	(11.1/16.5)	31.0
R	6.5/11	280.3	55.3	16.3	16	54	36.9	172.1	(16.5/23.2)	35.7
Comb.	17 /29	272.7	52.3	9.2	16	46	32.8	178.9	(8.7/12.6)	32.9
N + R **	17 /29	274.9	53.2			47	34.0	177.0		33.8
Griesbachian – Dienerian — no weighting										
N	18	267.4	49.2	9.1	15	47	29.2	183.6	(8.0/12.0)	30.1
R	11	277.4	56.2	11.0	18	52	37.4	175.2	(11.4/15.8)	36.7
Comb.	29	270.9	52.0	6.9	16	49	32.2	180.7	(6.4/9.4)	32.6
N + R	29	272.3	53.0			49	33.4	179.5		33.6

All-site means

	N	Mean direction				VGP				Lat. (°N)
		Decl. (°)	Incl. (°)	α_{95} (°)	K	Lat. (°N)	Long. (°E)	α_{95} (°)	K	
Gries.–Diener.	2	282.2	55.0	20.3	153	36.9	173.5	17.1	201	35.5
Smith.–Spath.	2	313.8	57.6	17.4	208	44.5	145.1	25.0	101	38.2
Early Triassic										
All	3	295.5	57.1	16.5	57	41.0	161.4	18.9	44	37.7

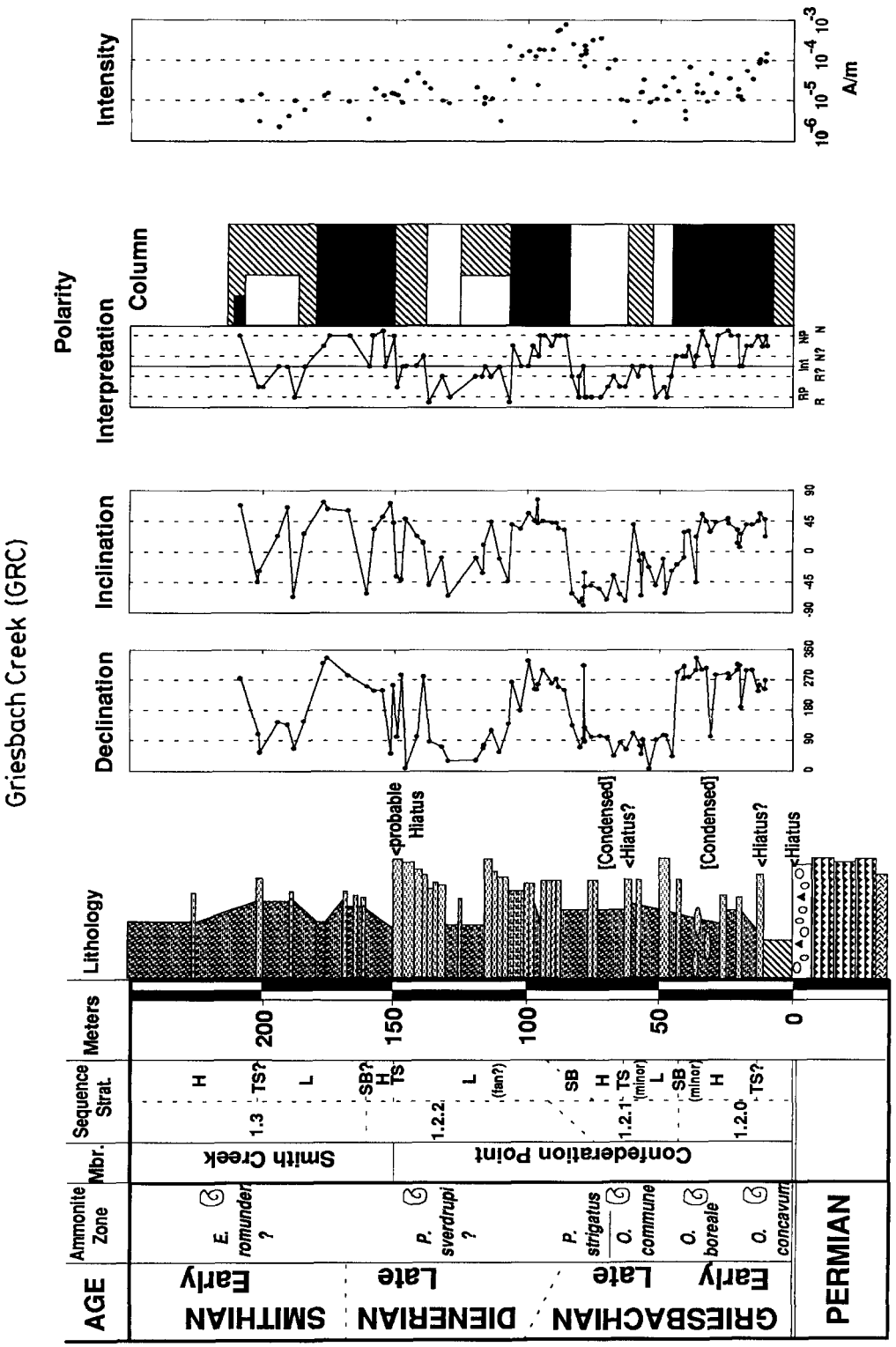


Fig. 5. For caption, see p. 82.

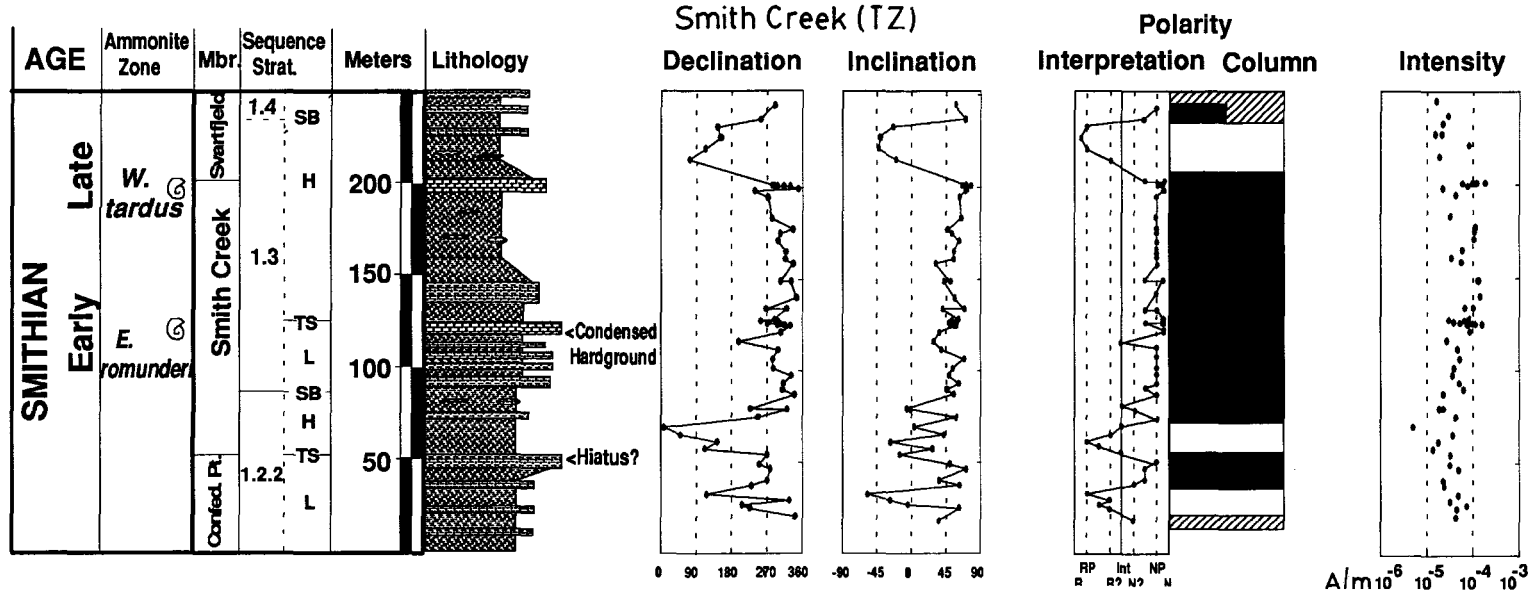


Fig. 5. For caption, see p. 82.

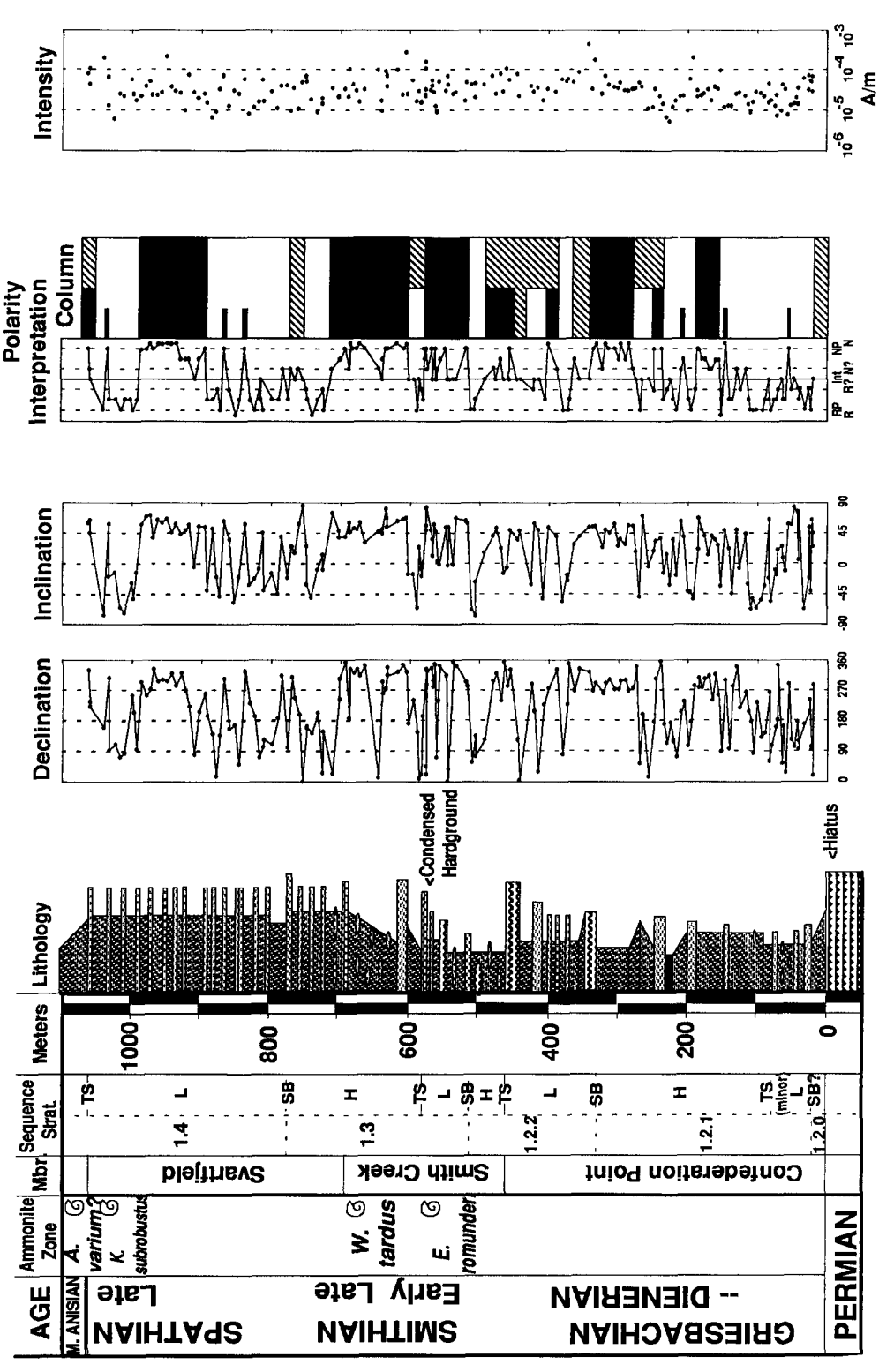


Fig. 5. Magnetostratigraphic sections with biostratigraphy (level of ammonite horizons in each section are indicated), lithostratigraphy, depositional sequence interpretation (nomenclature following Haq et al. [41,42]), characteristic directions of magnetization (declination, inclination, and intensity), and polarity interpretation (with an index of the reliability of the characteristic direction). In the sequence-stratigraphy column, *TS* = transgressive surface, *H* = transgressive or highstand deposits, *SB* = sequence boundary and *L* = regressive or lowstand deposits. In the polarity column, black is normal polarity.

been about 30° of clockwise rotation of the sites with respect to the ancient north magnetic pole (Fig. 4). This rotation is indicated both by section MB and by comparing all sites. This clockwise rotation is further supported by the position of the Smithian age direction from section TZ along the trend between the mean directions of Griesbachian–Dienerian and of Smithian-Spathian (Table 1). The rapid apparent polar wander seen by these sites during the Early Triassic requires further confirmation from other North American localities. Paleomagnetic results from different members of the Lower Triassic Moenkopi Formation, which primarily spans Smithian to Anisian, displayed no apparent polar wander [19]; therefore the episode of rapid apparent polar wander indicated by this Arctic data probably took place mainly during the Griesbachian to late Smithian.

(4) The mean inclination at all sites is 57°, indicating a paleolatitude of 37° N. There is an indication that a small amount of northward drift accompanied the rotation of the Arctic sites through the Early Triassic, but the amount of paleolatitude change falls within the uncertainties of the pole positions. It is possible that these clay-rich clastic sediments may have experienced some post-depositional shallowing of inclinations (as suggested by Tauxe and Clark [62] for nearby early Tertiary sites on Ellesmere Island). However, the mean inclination of 57° for the Arctic Early Triassic directions is identical to that predicted by the Early Triassic cratonic pole from North America [47], indicating that inclination error is not a significant factor.

(5) Comparison of the Smithian–Spathian mean pole of these Arctic sites (145.1° E, 44.5° N; α_{95} = 25.0°) with the mean Early Triassic pole of North America (118.3° E, 46.0° N) [63], which is mainly derived from formations of Smithian–Spathian age, implies that these sites have rotated about 23° counterclockwise with respect to cratonic North America. This cratonic pole for North America compensates for the clockwise rotation of the Colorado Plateau [63–65] as evident from the pole from the Lower Triassic Moenkopi Formation of 103° E, 55° N ($dp/dm = 1.6°/3.2°$) [19].

Early Permian paleomagnetism of the nearby Esayoo Basalt on Ellesmere Island yielded a pole (159.2° E, 48.1° N, $\alpha_{95} = 6.1°$) [66] indicating a similar 25° ± 12° of counterclockwise rotation relative to the mean Early Permian pole from cratonic eastern North America (129° E, 41° N, $\alpha_{95} = 8°$) [67]. Directions from mid-Cretaceous volcanic sills and dikes intruding the Triassic sediments in this region yield a pole (203° E, 67° N, $\alpha_{95} = 6.6°$) [55], indicating only about 12° ± 10° of counterclockwise rotation relative to the mean Cretaceous pole from cratonic North America (196° E, 71° N, $\alpha_{95} = 4.9°$) [69]. This counterclockwise rotation of this Arctic region has been attributed to the combined effects of Ellesmere Island moving with Greenland during the opening of the Labrador Sea and of later local block rotations in response to sinistral shear as Greenland drifted northeast during the formation of Baffin Bay [55,66]. The large error limits on these Permian, Triassic and Cretaceous poles do not allow determination of the timing of this counterclockwise rotation.

5. Magnetic stratigraphy

A stratigraphic plot was prepared for each section displaying directions and intensities of characteristic directions of magnetization and the associated polarity interpretations and quality rating with respect to lithology, depositional sequence interpretations, and ammonite zonations (Fig. 5). Polarity zones, consisting of clusters of samples having similar polarity, are shown in a generalized polarity column. These polarity columns generally ignore samples having questionable or indeterminate polarity. "Orphan" samples, having polarity interpretations opposite those of adjacent blocks of samples, are represented by a partial bar. Intervals lacking samples of definitive polarity are indicated by diagonal shading—a feature particularly common in section GRC.

The main features of the magnetostratigraphies of the three sections can be correlated using biostratigraphy, lithostratigraphy and depositional sequence facies patterns as guides (Fig. 6). In this correlation diagram, variations in relative sedimentation rates among the sections were partially adjusted in order to facilitate visual comparison and correlation of the magnetic polarity patterns. Section TZ is plotted at 150% of the MB scale in order to equate the stratigraphic distance between marker beds of the Smith Creek Member. Section GRC is plotted at 300% of the MB scale to equate

the relative thicknesses of the Confederation Point Member.

For reference purposes, individual polarity zones have been assigned a systematic nomenclature: lettered according to the geologic stage in which they occur, assigned *N* or *R* to denote polarity, and numbered consecutively from the base of each stage; for example, DN1 denotes the lowest observed normal-polarity zone in the composite polarity scale for the Dienerian stage. Due to relatively wide spacing of ammonite-rich horizons and lack of a precise "boundary stratotype", the base of the Spathian and Smithian stages are tentatively assigned to the level of depositional sequence boundaries (SB in the diagrams) at the beginning and end of sequence UAA-1.3 respectively, and the base of the Dienerian stage was placed between the appropriate ammonite-bearing levels of section GRC. These stage boundary assignments are only for purposes of polarity zone nomenclature, and are not to be considered as recommendations for international boundary stratotypes. Accordingly, the associated age nomenclature for polarity zones is also considered to be provisional. This uncertainty in stage boundaries is especially great for the Dienerian portion.

The main features of the magnetic polarity patterns of the different sections can then be compiled into a composite scale of the magnetic polarity pattern, ammonite zonation and depositional sequences for the Arctic stratotypes of the Early Triassic stages (Fig. 7). The relative thickness and features of each ammonite zone and polarity zone are plotted according to the adjusted stratigraphic columns (Fig. 6).

The Griesbachian stage consists of normal polarity (polarity zone GN1) for its lower two ammonite zones (*Otoceras concavum* and *Otoceras boreale* zones); its upper two ammonite zones (*Ophiceras commune* and *Pachyproptychites strigatus* zones) appear to consist predominantly of reversed polarity zones GR1–GR2. The polarity interpretation is based primarily upon the Griesbachian sequence at section GRC (Fig. 5), in which all the Griesbachian ammonite zones can be recognized. It is probable that the transgressive surface forming the sharp boundary between the *O. commune* zone and overlying *P. strigatus* zone is equivalent to the transgressive surface terminating the lowest prograding sequence of section MB (at about the 80 m level of MB). Therefore, the reversed-polarity zone comprising the lowermost beds at MB is also polarity zone GR1. The expanded transgressive to highstand portion of depositional sequence UAA-1.2.1 at section MB indicates the presence of a normal-polarity zone GN2. This polarity zone is absent in the deeper-water facies of section GRC because hiatuses in clastic sedimentation produced during rising sea-levels display relatively longer durations in more basinal settings [42]. The slope-fan facies of the Griesbach Creek section receives the majority of its sedimentation during regressive phases or lowstands of sea level, whereas the outer shelf facies of the Ellesmere sections consists mainly of clastic progradation onto the shelf during extended highstands and initial phases of falling sea level.

The Dienerian portion of the upper Confederation Point Member did not provide a definitive magnetostratigraphy in any of the sections. Normal-polarity zone DN1 occurs at the depositional sequence boundary between UAA-1.2.1 and UAA-1.2.2, as defined by the lowest prograding siltstones in the middle of the Confederation Point Member at sections MB and GRC, and is tentatively assigned a Dienerian age, although ammonite index fossils for early Dienerian are absent. Relative thicknesses of polarity zones DR1, DN2 and DR2 are based primarily upon the results of the section TZ; within section MB these polarity zones are probably present, but not well defined. The transgressive surface boundary between the coarse-grained progradational lowstand facies of the upper Confederation Point Member and the overlying fine-grained highstand facies of the basal Smith Creek Member in section TZ coincides with a change from normal polarity (polarity zone DN2) to overlying reversed polarity (polarity zone DR2). As before, the more basinal facies of section GRC displays an extended duration to this hiatus, and polarity zones DN2 and DR2 are both absent.

Most of the overlying Smith Creek Member consists of normal polarity (polarity zone SmN1–SmN2). Section MB has the same general pattern as TZ. A transgressive surface underlies the condensed limestone-bed bearing the lowest *Euflemingites romunderi* ammonite marker of the early

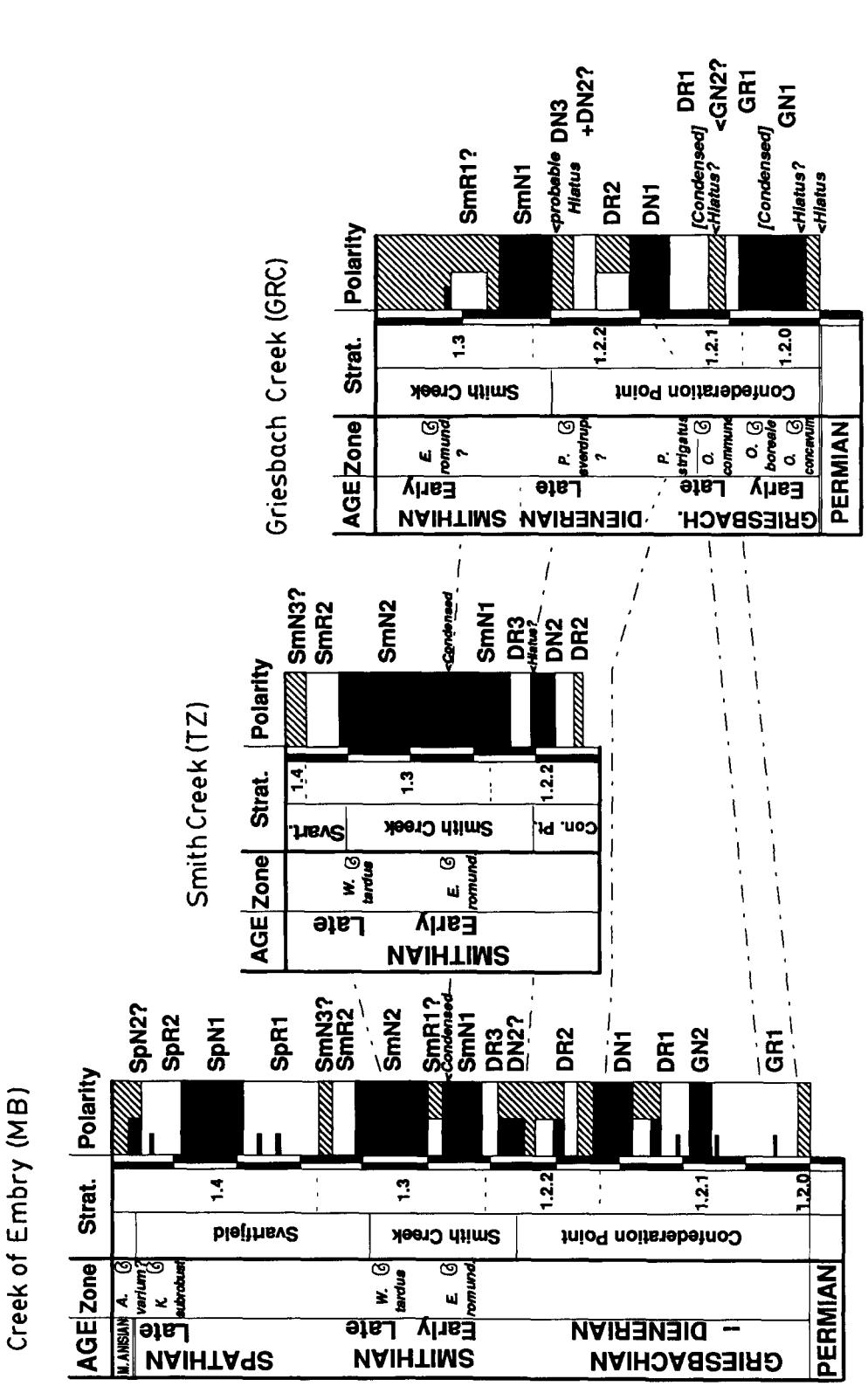


Fig. 6. Summary of all Lower Triassic magnetostratigraphic sections from the Canadian Arctic and the composite polarity column. Black is normal polarity; white is reversed polarity; diagonal shading indicates uncertain results or gaps in sampling. The composite polarity pattern (on the left side) indicates the relative abundance of normal vs. reversed polarity and the major polarity features observed for each ammonite zone and depositional sequence.

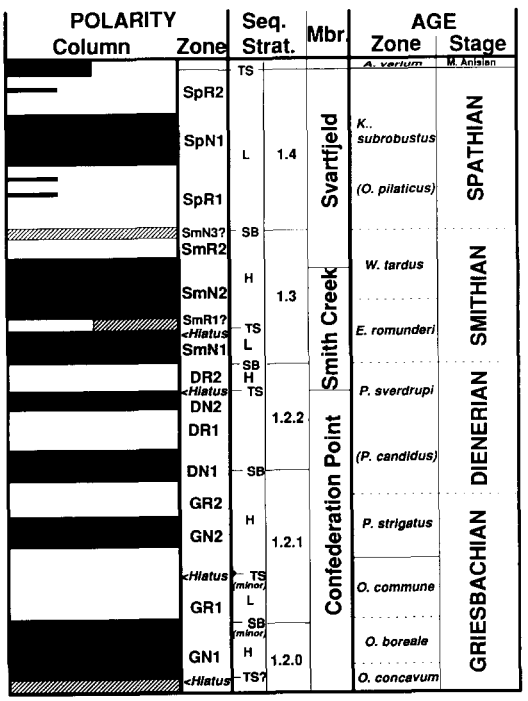


Fig. 7. Composite scale of magnetic polarity pattern and zones, biostratigraphy (ammonite zones and geologic stages), lithostratigraphy and sequence stratigraphy (depositional environments controlled by sea-level changes) for the Early Triassic, as derived from the Canadian Arctic sections. In the sequence-stratigraphy column, *TS* = transgressive surface, *H* = transgressive or high stand deposits, *SB* = sequence boundary and *L* = regressive or low stand deposits. In the polarity column, black is normal polarity and diagonal shading indicates uncertain results.

Smithian stage; therefore it is possible that a hiatus represents removed portions of the latest Dienerian and earliest Smithian stages. For this reason and the vague indication of a brief reversed-polarity zone SmR1? at sections MB and GRC (which has a higher sedimentation rate than TZ and therefore may be more complete), this extended interval of normal polarity of the Smith Creek Member is subdivided into polarity zones of early Smithian (SmN1) and of late Smithian (SmN2) age. The SmN1 polarity zone is correlated to the normal-polarity zone in the lower Smith Creek Member at GRC; but in the more basinal GRC section, it is probable that the transgressive surface and associated hiatus have removed much of the late Dienerian record (polarity zones DN2 and DR2). There may be a normal-polarity zone (SmN3?) at the sequence boundary defining the top of our Smithian stage.

The Spathian polarity pattern is recorded mainly by the MB section. The Spathian has two reversed-polarity zones (SpR1 and SpR2) separated by a normal-polarity zone (SpN1). The late Spathian (*Keyserlingites subrobustus* zone) may be partially absent at the transgressive boundary to Anisian sediments. The two samples of middle Anisian age are of normal polarity.

Correlations of the magnetostratigraphy of the three sections are not well defined in some intervals, particularly the upper portions of section GRC. This is probably due to a combination of poor magnetization and vague stratigraphic control. Hiatuses and transgressive condensation surfaces in these sections do not appear to have removed entire ammonite zones, but such intervals of non-deposition would distort the fine details of the polarity zonation. Further complications to comparing polarity patterns are the relatively high frequency of reversals and the inevitable distortions of polarity patterns in a clastic depositional environment with rising and falling sea-levels.

Therefore, this composite polarity column should be considered as a generalization of the polarity characteristics of the Early Triassic. The details and relative durations of the polarity pattern need to be verified within other Lower Triassic magnetostratigraphic sections containing either precise ammonite zonations or detailed depositional sequence determinations.

Unfortunately, other Lower Triassic sections which have detailed magnetostratigraphies lack either a precise ammonite zonation or a well-determined sequence stratigraphy. Redbed formations of North America and Europe have yielded reliable polarity patterns that can be regionally correlated [17–23], but their ages are poorly constrained. The continental facies of the Moenkopi Formation have Smithian and Spathian ammonites in some interbedded marine horizons, but these levels are too sparse to allow determination of ammonite zones. Nearshore marine sections in China have provided detailed magnetostratigraphies [29,30]; however, these sections lack ammonites above the basal condensed late Griesbachian [38]. All of these magnetostratigraphic sections appear to display approximately three to five polarity zones per geologic stage (estimated

age span)—the same reversal frequency as is recorded in the Arctic sections. Eventually, it may be possible to apply concepts of sequence stratigraphy to other sections in order to provide an alternative means of correlation control. No magnetostratigraphies have been published from Lower Triassic sections in the Salt Ranges or European Arctic which have sequence stratigraphy [42].

Another complication is that our Arctic sections represent a clastic outer shelf to submarine-fan environment, whereas most other published magnetostratigraphic sections have been in Lower Triassic continental or nearshore marine facies. As a result, low sea-level stands, corresponding to higher sedimentation rates in the Arctic sections, may be represented by exposure surfaces in coeval nearshore settings [42]. Conversely, slow accumulation of highstand facies in these Arctic sections would correspond to pulses of sedimentation in nearshore settings. This comparative "out-of-phase" in relative sedimentation rates between the Arctic and coeval nearshore sections would introduce considerable distortion in the magnetostratigraphic patterns of both environments, and hence preclude visual matching of polarity patterns without having independent biostratigraphic (preferably ammonite) age control and taking depositional sequence interpretations into consideration.

6. Conclusions

Magnetostratigraphic sampling of Lower Triassic sections in the Canadian Arctic, including the stratotypes for the Griesbachian and Smithian stages, have yielded an internally consistent polarity pattern which has been tied to ammonite zones. These polarity zones are recorded in clastic sediments of outer shelf to submarine-fan environments, which also display depositional sequences. The age and pattern of these depositional sequences appear to match the Early Triassic interval of the eustatic global sea-level curve of Haq et al. [41,42]. The polarity pattern has been tied to this sequence stratigraphy. Every magnetic polarity zone is recorded in at least two sections, with the exception of the lowest portion of the Griesbachian stage (present only in the stratotype) and the Spathian stage (extensively sampled in only one section). The average reversal frequency is approximately one pair of normal- and reversed-polarity zones per ammonite zone; if such ammonite zones have an average duration of 1 m.y. as is assumed for the Jurassic, then this corresponds to two magnetic reversals per 1 m.y.

The overall mean pole from these Lower Triassic sections is at 161.4° E, 41.0° N ($\alpha_{95} = 18.9°$). There is an indication of possible clockwise rotation between the Griesbachian–Dienerian and the Smithian–Spathian. The Smithian–Spathian pole indicates that the sites on Ellesmere and Axel Heiberg islands have undergone about 25° of counterclockwise motion with respect to cratonic North America, but with no change in latitude.

Correlation of the Arctic composite of ammonite zones, magnetostratigraphy and sequence stratigraphy (Fig. 7) to other Lower Triassic magnetostratigraphic studies is inhibited by lack of precise age control on most previous sections and by the predicted distortions of such a polarity pattern in the varying sedimentation rates in nearshore and continental environments during fluctuating sea levels. When this composite time scale has been independently verified in other magnetostratigraphic sections having either ammonite zonations and/or interpreted depositional sequences will then be possible to use it for global correlation of Lower Triassic strata and will provide a supplemental means of defining the stage boundaries of the Early Triassic.

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